

# Learning the Routing Policy for VIA-SD: RL-Gated Hierarchical Speculative Decoding

viasd-rl experiments

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## Abstract

We reimplement VIA-SD (arXiv:2606.12243) — speculative decoding with a slim, layer-skipped intermediate verifier  $q'$  — and replace its hand-tuned  $(\alpha_1, \alpha_2)$  confidence thresholds with a **learned per-token gating policy** trained by imitation and reinforcement learning (REINFORCE, GRPO, and a per-token-reward variant). On a GSM8K slice with a Qwen2.5 0.5B→14B drafter/verifier pair, the learned gate **dramatically outperforms the paper’s fixed thresholds** (accuracy 0.21–0.40 → 0.88–0.91) at fewer expensive verifier calls. We also report honest negative results: plain lossless speculative decoding remains Pareto-competitive at this model gap, and the slim-verifier middle tier only pays off when the gate is learned.

## 1 Methodology (outline)

- **Pipeline.** Drafter  $p$  (0.5B) proposes a  $\gamma$ -token block; for each token a gate routes to one of three actions: *accept*  $p$ ’s token (cheap), *regenerate* with the slim verifier  $q'$  (a layer-skipped copy of the full verifier  $q$ , medium), or *escalate* to the full  $q$  (expensive).  $q$  is Qwen2.5-14B.
- **Learned gate.** The fixed  $(\alpha_1, \alpha_2)$  thresholds are replaced by a tiny MLP over  $q$ -free features (drafter/ $q'$  entropy, confidence ratio, agreement, KL, position).  $q$  is used only at training time, never as a policy input.
- **Training.** (1) *Imitation*: an oracle that may consult  $q$  labels the cost-minimal route reproducing greedy- $q$ ; the MLP is fit with class-weighted cross-entropy. (2) *RL refinement*: REINFORCE (global baseline), GRPO (group-relative advantage), and a *v2 per-token* variant (per-token reward  $r_t = \text{match}_t - \lambda \text{cost}_t$  with per-token advantages and an overhead-corrected cost term).
- **Slim verifier  $q'$ .** Built by skipping  $\sim 45\%$  of  $q$ ’s layers. The skip mask is chosen by DIMR, an offline search minimizing the paper’s Eq. 11 margin-violation cost between  $q$  and  $q'$  (faithful metric;  $\sim 50\%$  lower than an evenly-spaced mask).
- **Metrics.**  $q$ -calls/token (full-verifier invocations per token; hardware-independent), GSM8K accuracy, and speedup. Speculative decoding’s win is *latency*, but measured wall-clock here is dominated by a  $\sim 10$  ms eager per-call overhead floor; we report an **overhead-free bandwidth-model speedup** (per-forward cost = active-params $\times 2$  bytes/HBM-bandwidth), validated against a measure-and-subtract estimate.

## 2 Best results

| Method                                | GSM8K acc    | q-calls/tok  | speedup (bw) |
|---------------------------------------|--------------|--------------|--------------|
| greedy- $q$ (reference)               | 0.887        | 1.000        | 1.00×        |
| plain SD (lossless)                   | 0.907        | 0.259        | 3.30×        |
| via_fixed (paper, tuned)              | 0.40*        | 0.31         | 1.76×        |
| via_imit (imitation only)             | 0.34         | 0.29         | 1.69×        |
| via_rl REINFORCE                      | 0.880        | 0.243        | —            |
| via_rl GRPO                           | 0.853        | <b>0.162</b> | —            |
| via_rl REINFORCE + DIMR               | <b>0.907</b> | 0.250        | 2.22×        |
| via_rl v2 per-token ( $\lambda=0.3$ ) | 0.713        | 0.104        | <b>3.53×</b> |

Table 1: Headline comparison (0.5B→14B, GSM8K). \*via\_fixed peaks at  $\approx 0.50$  even under a full 9-config threshold sweep. q-calls/token is exact and hardware-independent; bandwidth speedup is reported where measured. Accuracy uses  $n=80\text{--}150$  (see caveats).

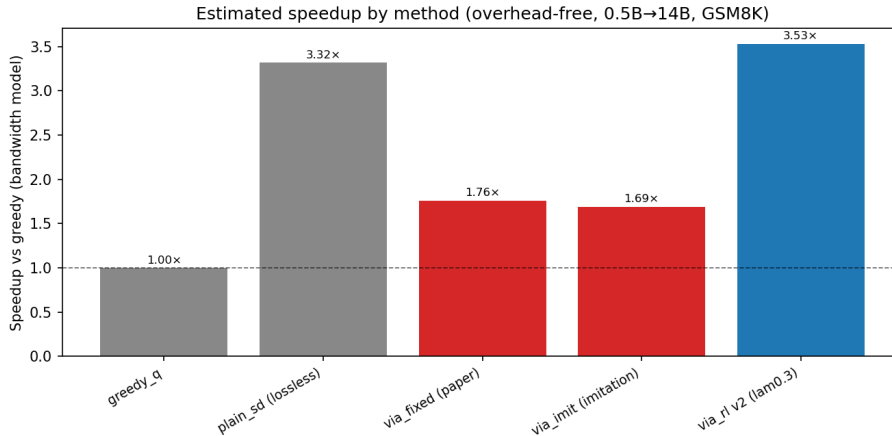


Figure 1: Overhead-free (bandwidth-model) speedup by method. The learned v2 gate and plain SD reach  $\sim 3.3\text{--}3.5\times$ ; the paper’s fixed-threshold gate is slow because it over-escalates to the full model.

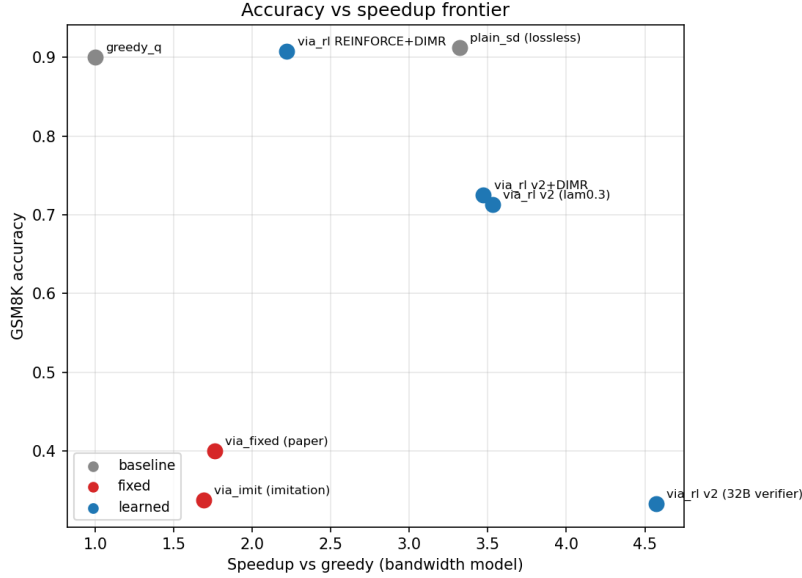


Figure 2: Accuracy vs. speedup. **Learned** gates dominate **fixed-threshold** ones. REINFORCE+DIMR matches plain SD’s accuracy (0.907) at lower speed (the  $q'$  tier overhead); the v2 per-token reward trades accuracy for speed.

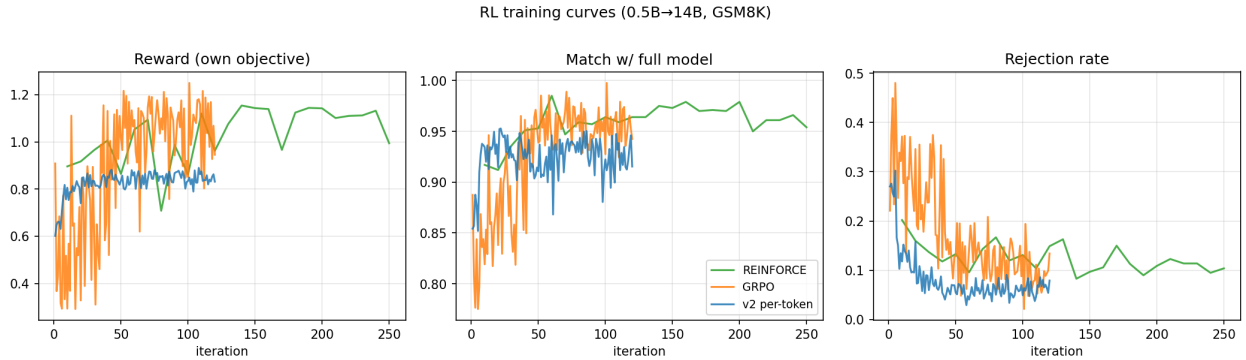


Figure 3: RL training curves. The v2 per-token reward (blue) is the lowest-variance and drives rejection lowest; GRPO (orange) is noisiest (group-level reward); REINFORCE (green) is intermediate. All converge to match  $\approx 0.95$  with the full model.

### 3 Findings

1. **Learned gate  $\gg$  fixed thresholds.** `via_ri` reaches 0.88–0.91 accuracy vs. `via_fixed`’s 0.21–0.40 (and  $\leq 0.50$  even fully tuned), at lower  $q$ -calls/token. This is the central result and is robust across masks and capacity gaps.
2. **RL fixes imitation’s exposure bias.** Imitation alone is 0.33; RL lifts it to 0.88+ by training on the policy’s own rollout distribution.
3. **DIMR helps only the learned gate.** A faithful, better  $q'$  lifts `via_ri` 0.88  $\rightarrow$  0.91 but does

not rescue via `fixed` — the fixed rule is brittle regardless of  $q'$ .

4. **The v2 per-token reward unlocks VIA-SD > plain-SD *on speed*** (escalation drops below the break-even,  $3.53\times > 3.30\times$ ), but overshoots on accuracy (0.71) at  $\lambda=0.3$ ; lower  $\lambda$  trades back toward accuracy.
5. **Honest negatives.** (i) Plain lossless SD stays Pareto-competitive at this gap — the  $q'$  middle tier’s per-block overhead doesn’t pay off. (ii) A *bigger verifier* (32B) *hurts* an accept-heavy policy (accuracy 0.33) because the fixed small drafter is then relatively weaker; a bigger *drafter* is the indicated lever.

## 4 Caveats

Accuracy at  $n=24\text{--}50$  carries a  $\pm 0.06\text{--}0.10$  CI, so small accuracy gaps are within noise; the large gaps (learned vs. fixed) are not. Speedups are an overhead-free cost *model*, not a stopwatch on an optimized engine (CUDA-graphs/static-cache were incompatible with the dynamic KV-cache loop). Single task (GSM8K), single pair family (Qwen2.5).